

HOW TO ENSURE PRODUCT RELIABILITY

Electronic device reliability and repair is a major concern to consumers. Here are a few things that the product designers and sellers can do to keep consumers satisfied. Also given are buying and repair guidelines for consumers to get the best value for their money



■ DR S.S. VERMA

Electronic devices containing delicate components can be easily damaged, unless they are rugged. A truly durable electronic device or gadget is one that substantially reduces the failure and repair cost while maintaining its look, feel and functionality over its lifetime.

Electronic equipment reliability and repair is a major concern to consumers due to many reasons including mass production, incompetent manpower to repair, least concern of companies about the quality of the product, and absence of strict rules and guidelines to protect consumer rights.

To meet the challenge of reliability, there is a need of leading-edge information that is critical to the creation

of reliable electronic devices and materials, focus on interdisciplinary communication in reliability of state-of-the-art electronic devices and microsystems, and focus on the materials and processes used in the manufacture of electronic devices, and the interfaces and surfaces of these materials.

Here are some guidelines for product manufacturers to ensure reliability of their products, obligations on part of the seller, role of the government in consumer protection, and guidelines for the consumers to get the best value for their money.

Guidelines for designers

Designers of electronic devices must be aware of the various points of failure and how to remove these through a sound design. Unlike mechanical parts,

electronic components generally do not wear out *per se*, but discrete analogue component parameters tend to drift over time and can cause problems with sensitive designs. Integrated circuits can undergo electro-migration. Furthermore, environmental effects, corrosion, vibration and temperature are of extreme concern. Transient stresses such as electrostatic discharge and lightning can also cause failures.

So to ensure that an electronic system will not fail beyond some tolerance, careful design principles have to be employed. The reliability of electronic devices can further be ascertained with proper warranty and repair services.

Government's duty

Laws should protect the consumers by regulating service dealers and service contract providers. These should also provide tips for consumers seeking repairs, as well as tips on purchasing electronic devices. To protect consumers, electronic devices, gadgets and major home appliances used or sold for personal, family, household or office use should be covered under the law for installation, repair, service and maintenance.

On part of the distributor

Almost all major electronic product distributors offer the original purchaser the same warranty as provided by the product manufacturer.

Basic electronic systems are warranted for one year on parts and labour. Warranty is limited only to